SAGAR PUBLIC SCHOOL

Integrated Programme • Topic: Gravitation

PRACTICE SHEET – 2

Gravitational Field due to Ring, Sphere and Shell; Applications

Name: _____	Class / Section: _____
Date: _____	Roll No.: _____
Time Allowed: 60 Mins	**Total Questions: 20**

Instructions: Attempt all questions. Section A has only one correct option; Section B may have more than one correct option; Section C needs full working. Time Allowed: 60 Mins.

Section A – Single Correct Option (Q.1 to Q.10)

Q.1 Let ω be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles. An object weighed at the equator gives the same reading as a reading taken at a depth d below earth's surface at a pole ($d \ll R$). The value of d is

(A) $\dfrac{\omega^2 R^2}{g}$ (B) $\dfrac{\omega^2 R^2}{2g}$ (C) $\dfrac{2\omega^2 R^2}{g}$ (D) $\dfrac{\sqrt{Rg}}{g}$

[Motion, Ex-I Q.3]

Q.2 A man of mass m starts falling towards a planet of mass M and radius R. As he reaches near to the surface, he realizes that he will pass through a small hole in the planet. As he enters the hole, he sees that the planet is really made of two pieces a spherical shell of negligible thickness of mass $2M/3$ and a point mass $M/3$ at the centre. Change in the force of gravity experienced by the man is

(A) $\dfrac{2}{3}\dfrac{GMm}{R^2}$ (B) 0 (C) $\dfrac{1}{3}\dfrac{GMm}{R^2}$ (D) $\dfrac{4}{3}\dfrac{GMm}{R^2}$

[Motion, Ex-I Q.5]

Q.3 The escape velocity for a planet is v_e. A tunnel is dug along a diameter of the planet and a small body is dropped into it at the surface. When the body reaches the centre of the planet, its speed will be

(A) v_e (B) $\dfrac{v_e}{\sqrt{2}}$ (C) $\dfrac{v_e}{2}$ (D) zero

[Motion, Ex-I Q.10]

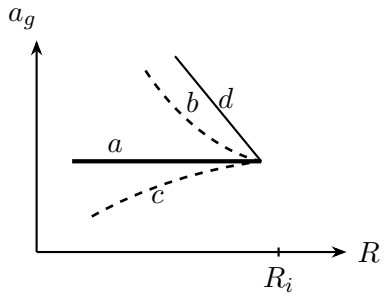
Q.4 If a tunnel is cut at any orientation through earth, then a ball released from one end will reach the other end in time (neglect earth rotation)

(A) 84.6 minutes (B) 42.3 minutes

(C) 8 minutes (D) depends on orientation

[Motion, Ex-I Q.12]

Q.5 A (nonrotating) star collapses onto itself from an initial radius R_i with its mass remaining unchanged. Which curve in figure best gives the gravitational acceleration a_g on the surface of the star as a function of the radius of the star during the collapse?



(A) a (B) b (C) c (D) d

[Motion, Ex-I Q.13]

Q.6 Select the correct choice(s):

(A) The gravitational field inside a spherical cavity, within a spherical planet must be non zero and uniform.
(B) When a body is projected horizontally at an appreciable large height above the earth, with a velocity less than for a circular orbit, it will fall to the earth along a parabolic path.
(C) A body of zero total mechanical energy placed in a gravitational field will escape the field.
(D) Earth's satellite must be in equatorial plane.

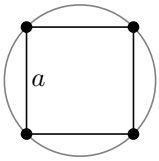
[Motion, Ex-I Q.20]

Q.7 Two planets A and B have the same material density. If the radius of A is twice that of B, then the ratio of the escape velocity $\frac{v_A}{v_B}$ is

(A) 2 (B) $\sqrt{2}$ (C) $1/\sqrt{2}$ (D) 1/2

[Nucleus, O-1 Q.13]

Q.8 Four identical particles of mass M are located at the corners of a square of side 'a'. What should be their speed if each of them revolves under the influence of others gravitational field in a circular orbit circumscribing the square?



(1) $1.41\sqrt{\frac{GM}{a}}$ (2) $1.21\sqrt{\frac{GM}{a}}$ (3) $1.35\sqrt{\frac{GM}{a}}$ (4) $1.16\sqrt{\frac{GM}{a}}$

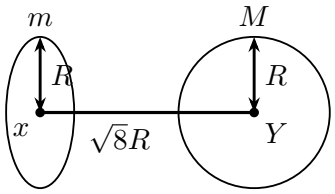
[Nucleus, JM Q.13 [JEE Main-2019]]

Q.9 Four identical particles of equal masses 1 kg made to move along the circumference of a circle of radius 1 m under the action of their own mutual gravitational attraction. The speed of each particle will be:

(1) $\sqrt{\frac{G}{2}\left(1+2\sqrt{2}\right)}$ (2) $\sqrt{G\left(1+2\sqrt{2}\right)}$ (3) $\sqrt{\frac{G}{2}\left(2\sqrt{2}-1\right)}$ (4) $\sqrt{\frac{\left(1+2\sqrt{2}\right)G}{2}}$

[Nucleus, JM Q.32 [JEE Main-2021]]

Q.10 Find the gravitational force of attraction between the ring and sphere as shown in the diagram, where the plane of the ring is perpendicular to the line joining the centres. If $\sqrt{8}\,R$ is the distance between the centres of a ring (of mass 'm') and a sphere (mass 'M') where both have equal radius 'R'.



(1) $\frac{\sqrt{8}}{9}\frac{GmM}{R}$ (2) $\frac{2\sqrt{2}}{3}\frac{GmM}{R^2}$ (3) $\frac{1}{3\sqrt{8}}\frac{GmM}{R^2}$ (4) $\frac{\sqrt{8}}{27}\frac{GmM}{R^2}$

[Nucleus, JM Q.39 [JEE Main-2021]]

Section B – One or More Than One Option Correct (Q.11)

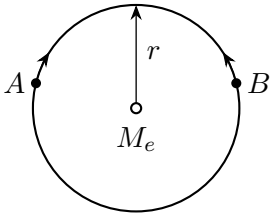
Q.11 Two masses m_1 and m_2 ($m_1 < m_2$) are released from rest from a finite distance. They start under their mutual gravitational attraction

(A) acceleration of m_1 is more than that of m_2
(B) acceleration of m_2 is more than that of m_1
(C) centre of mass of system will remain at rest in all the references frame
(D) total energy of system remains constant

[Motion, Ex-II Q.2]

Section C – Subjective / Numerical Problems (Q.12 to Q.20)

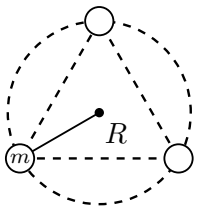
Q.12 Consider two satellites A and B of equal mass m, moving in the same circular orbit of radius r around the earth E but in opposite sense of rotation and therefore on a collision course (see figure).



(a) In terms of G, M_e, m and r find the total mechanical energy $E_A + E_B$ of the two satellite plus earth system before collision.
(b) If the collision is completely inelastic so that wreckage remains as one piece of tangled material (mass $= 2m$), find the total mechanical energy immediately after collision.
(c) Describe the subsequent motion of the wreckage.

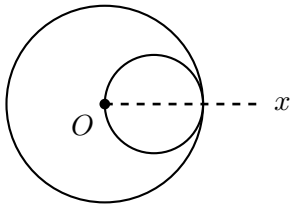
[Motion, Ex-III Q.9]

Q.13 A hypothetical planet of mass M has three moons each of equal mass 'm' each revolving in the same circular orbit of radius R. The masses are equally spaced and thus form an equilateral triangle. Find
(i) the total P.E. of the system
(ii) the orbital speed of each moon such that they maintain this configuration.



[Motion, Ex-III Q.13]

Q.14 A sphere of radius R has its centre at the origin. It has a uniform mass density ρ_0 except that there is a spherical hole of radius $r = R/2$ whose centre is at $x = R/2$ as in fig.
(a) Find gravitational field at points on the axis for $x > R$.
(b) Show that the gravitational field inside the hole is uniform, find its magnitude and direction.



[Motion, Ex-III Q.15]

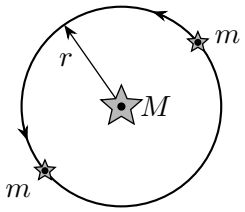
Q.15 A body moving radially away from a planet of mass M, when at distance r from planet, explodes in such a way that two of its many fragments move in mutually perpendicular circular orbits around the planet. What will be
(a) then velocity in circular orbits.
(b) maximum distance between the two fragments before collision and
(c) magnitude of their relative velocity just before they collide.

[Motion, Ex-III Q.16]

Q.16 A thin spherical shell of total mass M and radius R is held fixed. There is a small hole in the shell. A mass m is released from rest a distance R from the hole along a line that passes through the hole and also through the centre of the shell. This mass subsequently moves under the gravitational force of the shell. How long does the mass take to travel from the hole to the point diametrically opposite.

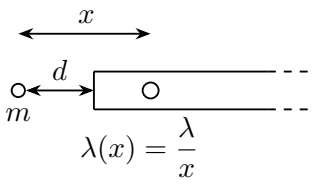
[Motion, Ex-III Q.17]

Q.17 A certain triple-star system consists of two stars, each of mass m, revolving about a central star, mass M, in the same circular orbit. The two stars stay at opposite ends of a diameter of the circular orbit, see figure. Derive an expression for the period of revolution of the stars; the radius of the orbit is r.



[Motion, Ex-IV Q.2]

Q.18 Find the gravitational force of interaction between the mass m and an infinite rod of varying mass density λ such that $\lambda(x) = \lambda/x$, where x is the distance from mass m. Given that mass m is placed at a distance d from the end of the rod on its axis as shown in figure.



[Motion, Ex-IV Q.3]

Q.19 A ring of radius R is made from a thin wire of radius r. If ρ is the density of wire then what will be the gravitational force exerted by the ring on the material particle of mass m placed on the axis of ring at a distance x from its centre. Show that the force will be maximum when $x = R/\sqrt{2}$ and the maximum value of force will be given as $F_{\max} = \dfrac{4\pi^2 G r^2 \rho m}{(3)^{3/2} R}$.

[Motion, Ex-IV Q.5]

Q.20 A small body of mass m is projected with a velocity just sufficient to make it reach from the surface of a planet (of radius $2R$ and mass $3M$) to the surface of another planet (of radius R and mass M). The distance between the centers of the two spherical planets is $6R$. The distance of the body from the center of bigger planet is 'x' at any moment. During the journey, find the distance x where the speed of the body is (a) maximum (b) minimum. Assume motion of body along the line joining centres of planets.

[Nucleus, S-1 Q.2]

ANSWER KEY – Practice Sheet 2

Answers are taken from the source answer keys (Motion: Answer Ex-I to Ex-IV; Nucleus: Answer Key page). JM answers are option numbers.

Q.	Source	Answer
1	Motion Ex-I Q.3	A
2	Motion Ex-I Q.5	A
3	Motion Ex-I Q.10	B
4	Motion Ex-I Q.12	B
5	Motion Ex-I Q.13	B
6	Motion Ex-I Q.20	C
7	Nucleus O-1 Q.13	A
8	Nucleus JM Q.13	(4)
9	Nucleus JM Q.32	(4) (options (1) and (4) are printed identically in the source; the key gives (4))
10	Nucleus JM Q.39	(4)
11	Motion Ex-II Q.2	A, D
12	Motion Ex-III Q.9	(a) $-\dfrac{GmM_e}{r}$; (b) $-\dfrac{2GmM_e}{r}$; (c) not given in the source key
13	Motion Ex-III Q.13	(i) $-\dfrac{3Gm}{R}\left(\dfrac{m}{\sqrt{3}}+M\right)$; (ii) $\sqrt{\dfrac{G}{R}\left(\dfrac{m}{\sqrt{3}}+M\right)}$
14	Motion Ex-III Q.15	(a) $\vec{g}=+\dfrac{\pi G\rho_0 R^3}{6}\left[\dfrac{1}{\left(x-\frac{R}{2}\right)^2}-\dfrac{8}{x^2}\right]\hat{i}$; (b) $\vec{g}=-\dfrac{2\pi G\rho_0 R}{3}\,\hat{i}$
15	Motion Ex-III Q.16	(a) $\sqrt{\dfrac{GM}{r}}$; (b) $r\sqrt{2}$; (c) $\sqrt{\dfrac{2GM}{r}}$
16	Motion Ex-III Q.17	$2\times\sqrt{\dfrac{R^3}{GM}}$
17	Motion Ex-IV Q.2	$\dfrac{4\pi r^{3/2}}{\sqrt{G(4M+m)}}$
18	Motion Ex-IV Q.3	$\dfrac{Gm\lambda}{2d^2}$
19	Motion Ex-IV Q.5	"Show that" question; the result is stated in the question (no separate answer in the source key)
20	Nucleus S-1 Q.2	(a) $x=2R$; (b) $x=3R\left(3-\sqrt{3}\right)$

SELF ASSESSMENT – Practice Sheet 2

Fill this in after checking your answers with the Answer Key.

Time Allowed: 60 Mins	**Time Taken:** _____ Mins
Start time: _____	End time: _____
Finished within the allowed time? ☐ Yes ☐ No (extra / saved time: _____ Mins)	

Total	Attempted	Correct	Incorrect	Not attempted
20				

Score (Correct / 20): _____ Percentage: _____ %

Question-wise tracker write ✓ (correct), ✗ (incorrect) or – (not attempted)

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Reason for the incorrect / unattempted questions (write question numbers)

Concept not clear: _____	Formula forgotten: _____
Calculation / silly mistake: _____	Ran out of time: _____

Personal feedback guide (tick the band that matches your score)

	Correct	What to do next
☐	18–20	Excellent. Concept and speed are both strong. Move to harder JEE-level problems and keep revising formulae once a week.
☐	14–17	Good. Re-do every incorrect question without looking at the solution, then note the formula or idea you missed.
☐	9–13	Fair. Revise the theory of fields, potential and orbits (shell, ring, sphere results), then re-attempt this sheet after 3 days.
☐	0–8	Needs attention. Re-read the chapter notes and solved examples first, then attempt this sheet again in two parts of 30 mins.

My personal feedback / action plan:

Teacher's remarks:

Student's signature: _____ Teacher's signature: _____